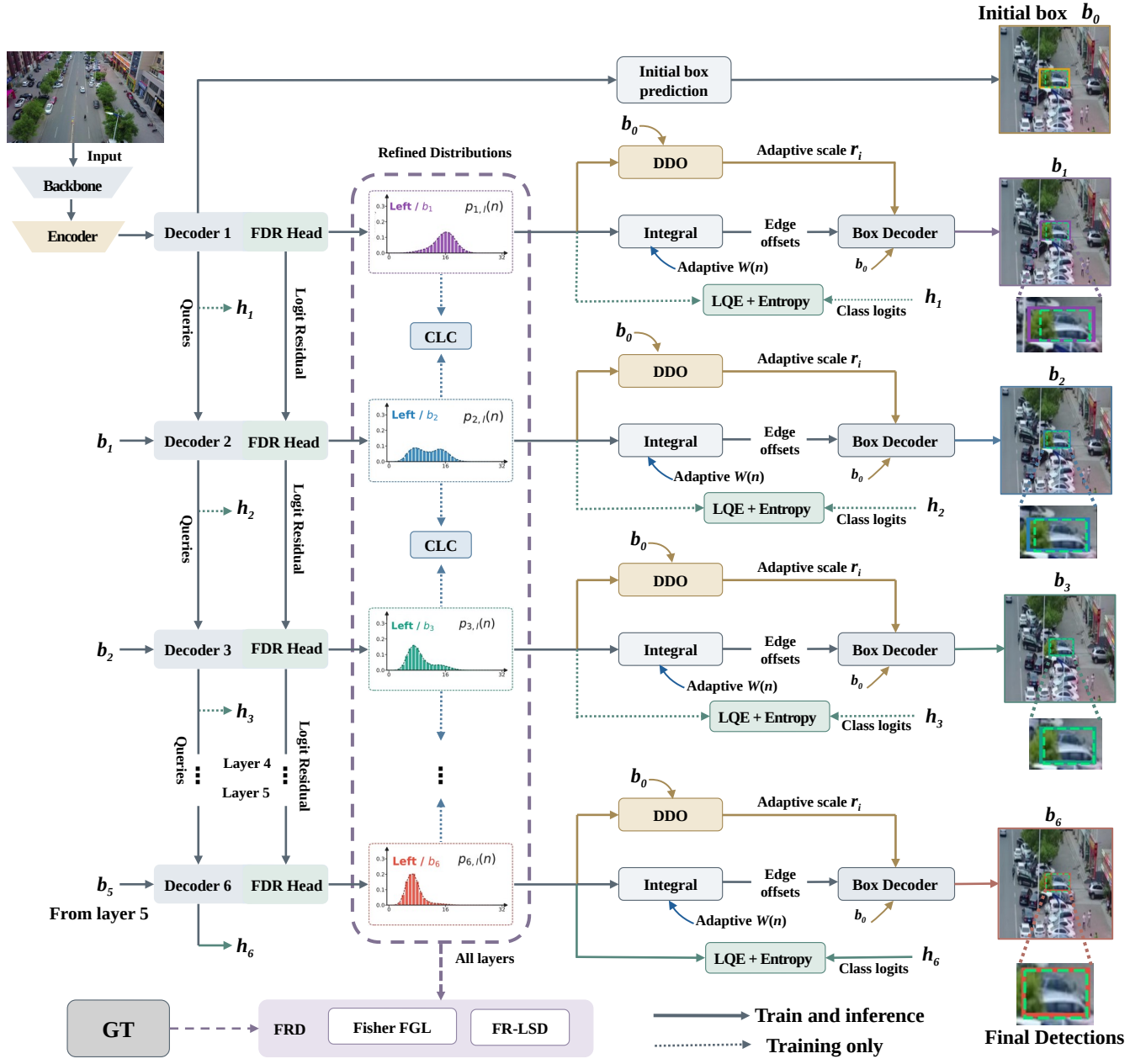
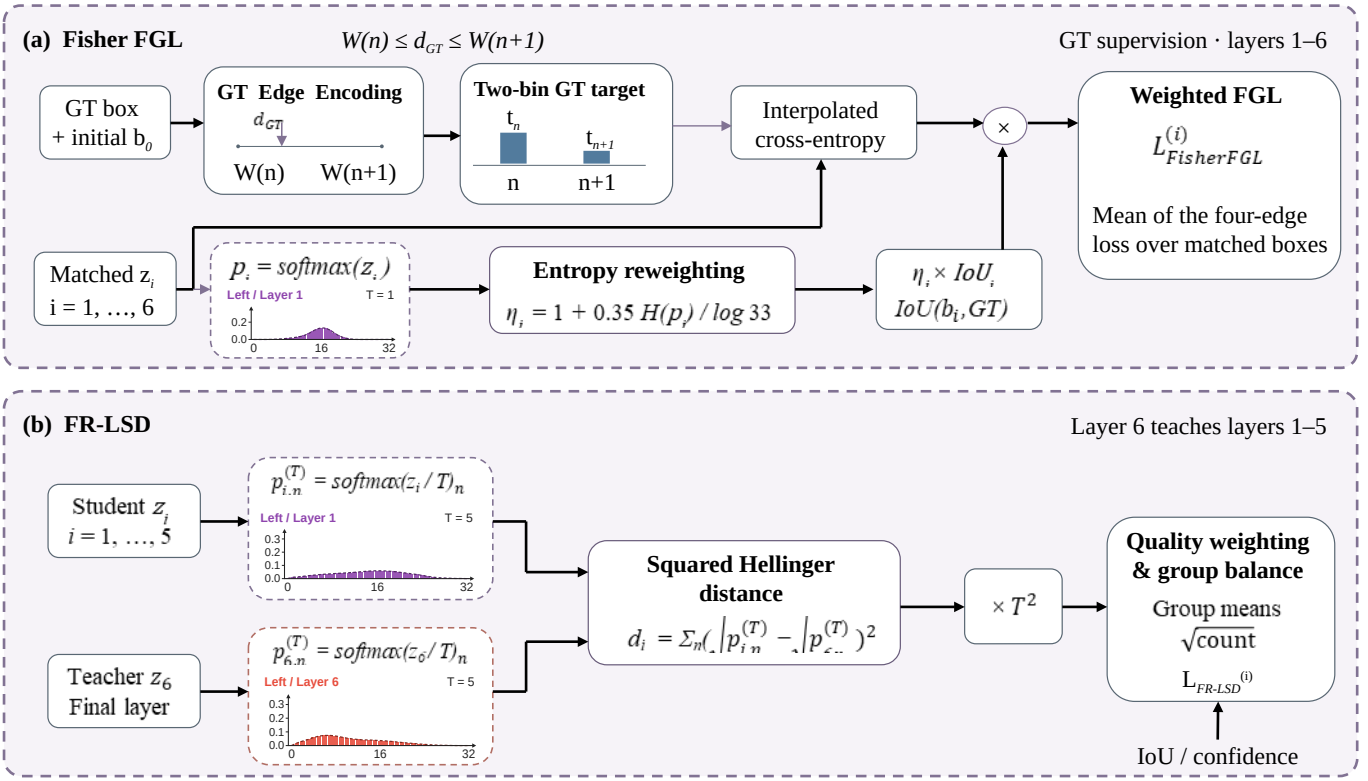
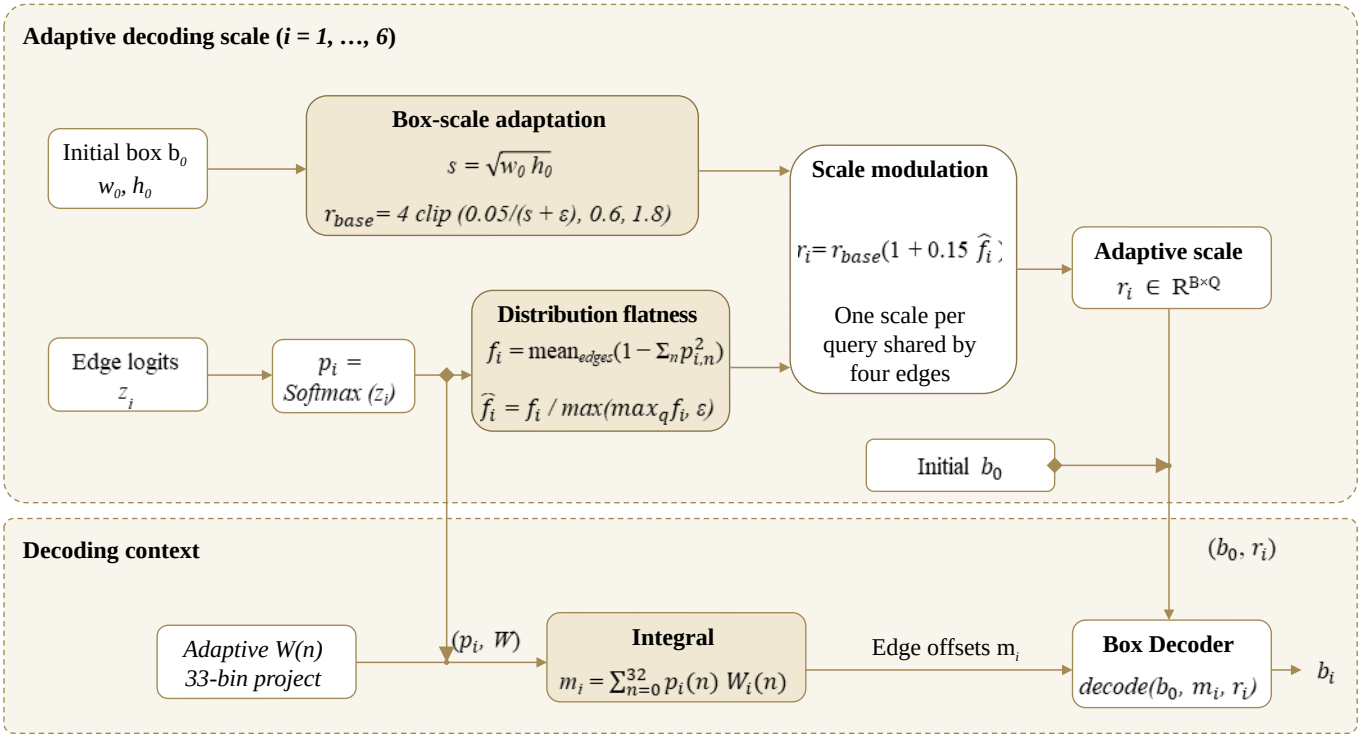
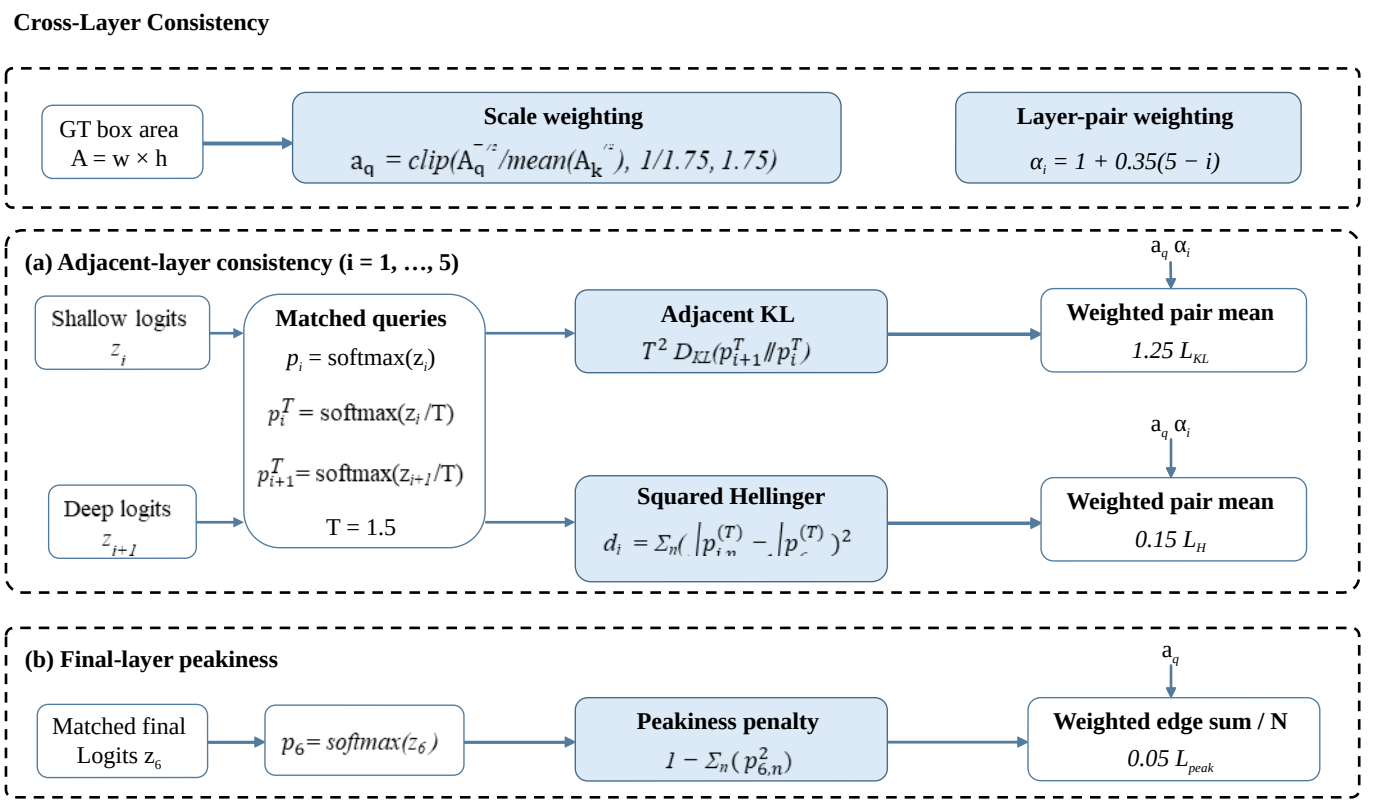

Initial box b_0
Input
Backbone
Encoder
Initial box prediction
Refined Distributions
Decoder 1
FDR Head
Queries
h_1
Logit Residual
b_1
Decoder 2
FDR Head
Queries
h_2
Logit Residual
b_2
Decoder 3
FDR Head
Queries
h_3
Layer 4
Layer 5
Logit Residual
b_5
From layer 5
Decoder 6
FDR Head
h_6
Left / b_1 $p_{1,i}(n)$
Left / b_2 $p_{2,i}(n)$
Left / b_3 $p_{3,i}(n)$
Left / b_6 $p_{6,i}(n)$
CLC
CLC
All layers
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_1
b_1
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_2
b_2
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_3
b_3
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_6
b_6
Final Detections
GT
FRD
Fisher FGL
FR-LSD
Train and inference
Training only

Fisher–Rao Distribution Loss



Decoder Distribution Optimizer (DDO)




Cross-Layer Consistency
GT box area
A = w × h
Scale weighting
Layer-pair weighting
α_i = 1 + 0.35(5 − i)
(a) Adjacent-layer consistency (i = 1, …, 5)
Shallow logits
Deep logits
Matched queries
p_i = softmax(z_i)
T = 1.5
Adjacent KL
Squared Hellinger
Weighted pair mean
1.25 L_KL
0.15 L_H
(b) Final-layer peakiness
Matched final
Logits z_6
p_6 = softmax(z_6)
Peakiness penalty
1 − Σ_n(p²_{6,n})
Weighted edge sum / N
0.05 L_peak

Localization Quality Estimator (LQE)

